

Badhrinarayanan K V

(412) 670-1355 | [Email](#) | [LinkedIn](#) | [Google Scholar](#) | [GitHub](#)

EDUCATION

Carnegie Mellon University

Pittsburgh, PA

M.S. Computer Vision GPA: 3.8/4.0

Aug. 2025 – Dec. 2026

- Relevant Coursework: Advanced Computer Vision, Visual Learning and Recognition, Learning for 3D Vision

Shiv Nadar University Chennai

Chennai, India

B.Tech Artificial Intelligence and Data Science GPA: 8.8/10

Aug. 2021 – Jul. 2025

- Relevant Coursework: Machine Learning, Deep Learning, Image and Video Processing, Natural Language Processing, Optimization Techniques

EXPERIENCE

Software Engineer Intern (Vision Pro Group)

May 2026 – Present

Apple

Cupertino, CA

- Developed a **Neural Shader** for **Apple Vision Pro**, rendering physically based materials in real time while achieving **35.1 PSNR** and **0.986 SSIM** against path traced data.
- Integrated **Image-Based Lighting (IBL)** and designed a **generalized neural shader model** capable of rendering unseen materials for robust **novel-view relighting**.
- Optimizing and deploying the model on the Vision Pro using Metal and presented the project to **Apple senior leadership**.

Graduate Research Intern

Jan 2026 – Present

Carnegie Mellon University

Pittsburgh, PA

- Developing an **adaptive Slot Attention** framework on **VGGT** for object-centric **3D scene understanding and segmentation**.

Undergraduate Research Intern

May 2024 – Aug. 2024

Indian Institute of Technology Madras

Chennai, India

- **3D Novel View Synthesis:** Co-developed a transformer-based, generalizable **Neural Radiance Field (NeRF)** for **lensless imaging**, enabling robust **3D scene reconstruction** from noisy optical inputs.
- Curated the **first multi-view lensless dataset**, achieving **22.8 PSNR** and outperforming **state-of-the-art (SOTA)** methods on *nerf-llff*, *ibrnet_collected*, and *real_iconic_noface* by **+2.6 PSNR**.

PUBLICATIONS

GANESH: Generalizable NeRF for Lensless Imaging • First Author • WACV 2025

PROJECTS

3D Gaussian Splatting Concept Transfer using Video Diffusion

- Developed a **3D Gaussian Splatting** framework for localized object editing, enabling realistic texture and material transfer while preserving scene geometry.
- Integrated a **Video Diffusion Model** to generate **360° multi-view consistent** edits with accurate texture alignment across novel viewpoints.
- Demonstrated high-fidelity style transfer on **Objaverse** assets while reducing geometric artifacts common in text-driven 3D editing methods.

Graph-RAG Enhanced Multimodal Medical VQA for Dermatology

- Developed **Med-GRIM**, a **Graph-RAG** medical VQA system integrating clinical knowledge with prompt-engineered SLMs, achieving **83.33%** VQA accuracy.
- Designed **BIND**, a multimodal encoder improving vision-language alignment, increasing embedding accuracy to **87.59%**.
- Built **DermaGraph**, a dermatology knowledge graph dataset enabling grounded multimodal retrieval for scalable medical visual question answering.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, R

Machine Learning: PyTorch, TensorFlow, NumPy, Pandas, OpenCV, Scikit-Learn, Keras, Neo4j, NLTK

Computer Vision: NeRF, Gaussian Splatting, Vision Transformers, Diffusion Models, Image Processing